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1 SCOPE AND AREA OF APPLICATION

This procedure is intended to govern the performance curve fit of new curves or the best fit of existing curves to address an unsatisfactory shape or if the fit must be revised.

2 ACRONYMS USED IN THIS DOCUMENT

AOR: Allowable operating region

BEP: Best efficiency point

ETA: Efficiency

FDE: Fluid Dynamics Engineer

HI: Hydraulic Institute

IQ: Intelliquip (pump selection software)

NFPA: National Fire Protection Association

POR: Preferred Operating Region

PRM: Proposal Manager

SSS: Suction specific speed

TE: Technical Engineer

TP: Technical Proposal

TPI: Trillium Pumps Italy

TPUS: Trillium Pumps US

3 DEFINITIONS

Software Tools

Molude 471 is used for generating new curves that with known specific speed (Ns) where at least one pump with “scaled” hydraulics was tested. Molude 471 shall also be used to scale and generate the curves for the new size using affinities laws for scaling and trimming. For curves already deployed or other conditions defined herein, other tools tailored to the task (e.g. excel calculations) may be employed provided they are consistent with this standard.

Derates

Fitting to H-Q and efficiency data shall be done in a neutral manner based on the test curves; no derates to provide testing margins shall be applied. Derates to address various test tolerance shall be applied in IQ and are not in the scope of this document.

The only changes in curve data input allowed are modifications due to some condition applied to the tested pump like coating, temperature (if test is done with hot oil), or speed scaling to the near synchronous speed (for tests with VFD).

For vertical pumps, if it is not possible to test the bowl performance, the pump performance should be corrected according to the estimated friction losses. It is known that for many tests done in the past not all information is available, and in those cases, friction losses to calculate the bowl performance can be ignored (since it is a conservative position).

NPSHr curves shall always be based on a fit to NPSH3 data.

If the NPSHr fitting is based on a single test and no other comparable data is available, a correction factor for test and manufacturing variation must be included to insure test passage. Curve fitting errors are usually negligible, but these should be verified and accounted for in the adjustment. Standard NPSH3 adjustment factors shall be 1.1x for sand cast impellers and 1.07x for investment cast unless a sampling of multiple tests indicates that lesser values can be used.

Where multiple tests or other comparable (ie scalable) data is available, NPSHr is aligned to the most conservative set of data available, with this rules order of importance:

1. In the POR, NPSHr shall be aligned to the most conservative set of data, with no additional correction.
2. In the POR, if there is no consistent data, we align to the worst test among all sizes of the same hydraulic.
3. NPSHr less than 1,5 m is not easy to measure due to test loop vacuum capability. Therefore, NPSHr less than 0.7 m in the POR is not published since it cannot be measured.
4. A geometric scale factor less than 1.0 could reduce the SSS. NPSHr curves lower than expected are not considered unless confirmed by several tests (minimum 3 with different materials).
5. If there is one test at size and it is aligned to the expected SSS from many tests of different sizes of the same hydraulic at the same speed, we use the test data with no correction.
6. If there is no test at size but larger and smaller sizes of the same hydraulic have the same SSS with aligned data, we use the theoretical scaled points.

Additional increases NPSHr to meet a suction specific speed target or to have a safety margin for operation (for example 40000 h of impeller safe operation) shall be done in IQ.

Curves for Fitting

Curves for fitting include head-capacity, efficiency and NPSHr. The minimum set of data are 2 trims, at maximum and minimum diameters, but is common practice to have at least 1 or 2 additional intermediate trims when the minimum diameter is over the 5% of trimming. Curves for one or more trims and speeds may be fitted depending on availability of test data and intended tolerance grade for the fitted curves. More specific guidance is provided in the following sections.

Reference Condition

Curves shall be corrected for deviations from a reference condition including impeller running clearance, hydraulic roughness (excludes coating like Belzona or super polishing), fluid temperature, staging, speed, and standard tolerance grade. Reference conditions shall be defined by product line.

Axial flow with angle regulation instead of trimming is not included in this standard work.

NPSHr shall always be NPSH3.

Refer to SOP's 483 and 485, respectively for refiguring curves for changes in roughness and clearance.

4 PROCESS TO FIT CURVE

4.1. PROCESS: TRIM CURVE

Trim Curves

All curves are to be formatted according to Molude 471. In the module, the first sheet, "input data for model test" (model test), is the basis for all other curves that are scaled from it. This approach allows the management of all curves with a single input in the first sheet and some important inputs in the "generated" sheets like maximum diameter (to set the scale factor), speed, exponent for trimming law of flow, head, and NPSH3.

The first sheet is the location to do the best fit of the Q-H, Q-eta and NPSH3, using some best practices to have a best fit of the plotted curves:

1. Use 8 points for Q-H-Efficiency and 6 points for Q-NPSH.
2. Points shall be 1st 0% Q, 2nd to 5th a percentage from 15% to 85% BEP, 6Th BEP, 7th a percentage from 110% to 120% of BEP, the last a percentage from 130% to 150% of BEP (note: very low specific speed pumps are limited to 105-110% of BEP due to volute choking; points 7 and 8 should be defined accordingly).

3. Insert maximum diameter points: 8 for Q-H-efficiency and 6 points for Q-NPSH. The first NPSH point shall correspond to MCSF at the maximum diameter.
4. Determine the minimum diameter and insert one or 2 intermediate diameters if tests are available.
5. Use the 2 trimming law exponents for flow and head to interpolate the trims in terms of Q-H and the BEP position of trims. The exponents are determined based on a visual comparison of the calculated curve with the test data.
6. Modify the maximum diameter values of efficiency (8 control values in the polynomial fitting section) to shape the efficiency curve. Verify the desired maximum efficiency is achieved with the fit.
7. Modify only the BEP values of the trim curves for the best fit to the test data and define the efficiency loss versus trim as a quadratic polynomial curve (figure Eta-trims)
8. Insert 6 points for Q-NPSH at minimum diameter. The first NPSH point shall be the MCSF at the minimum diameter.
9. Use the interpolation spline to perform any additional iteration needed on the fits.

In each generated sheet, exponents for efficiency are calculated (1 each for reduced/increased speed) to enable the use of a single curve for the full speed range of the pump.

Usually, the exponents for the trimming law of flow and head are the same for all specific speeds (Ns) but these can be modified according to test data in each generated size (all other sheets).

For curve without trimming tests, the method to establish the exponents is described in SOP- 475 Trimming Coefficient Definition section.

If a satisfactory fit is not achievable using the methods above, relating to such issues as exceeding the constraints of the intended tolerance grade, fit errors large enough to adversely impact the competitiveness of the final curves, or inadequately capturing particular characteristics of the hydraulic (e.g. head curve saddles), modifications to the approach can be made, including: overriding values of the calculated trim curves with those of curves fitted directly to test data, the use of additional curve points, use of multiple interpolated spline segments, or the use of other calculation tools to suit the particular task, being mindful that the best practice is to use the minimum possible number of points and spline segments.

Fit errors shall be limited according to the following table:

	Curve Fit Tolerance - Full Diameter Curve (%)				Curve Fit Tolerance - Trim Curve (%)				
	H-Q		Efficiency		H-Q		Efficiency		Max Power
Test Tolerance	In POR	Outside POR	In POR	Outside POR	In POR	Outside POR	In POR	Outside POR	Only the EOC or HSO (high Ns*)
Unilateral tolerance, up to and including HI 1U (-0% for H-Q, Efficiency)	+/-1.0	+/-2.0	+/-0.25	+0.5/-1.0	+/-1.0	+/-2.0	+/-0.25	+0.5/-1.0	+4 -0
Bilateral tolerance, up to and including HI 1B	+/-1.0	+2 -3	+0.3 -1	+0.5 -3	+/-2.0	+2 -4	+0.5 -2	+0.5 -4	+9 -1

*90(metric)/4,500(US) or higher, for the purpose of this evaluation

Table 1. Curve fitting tolerances.

For trims, especially minimum diameter, it is worth differentiating by product line (by market) since some can have a larger test tolerance that allows the search of a best fit of a range of diameters. Product lines that are sold and tested at a high volume (larger database of tests resulting in higher confidence in the measurement) can also be differentiated.

As an example, the API H-Q tolerance of +/- 3% can be achieved even with a different diameter than what is indicated in the bidding curve.

The SKB interpolates between 2 diameters or 3 diameters, depending on the curve inputs to the SKB. Interpolation among 3 curves requires verification of not only the single diameter fit but also the consistency of the data with trimming. It's misleading to focus ONLY on the single diameter error since the SKB interpolates on a region. For this reason we need to allow a greater error on the single diameter with the benefit of greater smoothness and consistency of the correlation between diameters. To this end, a best fit of the H-Q, ETA-Q data is performed to have the best fit in the region bounded by Dmax, 5% trim and POR.

Example of Molude 471:

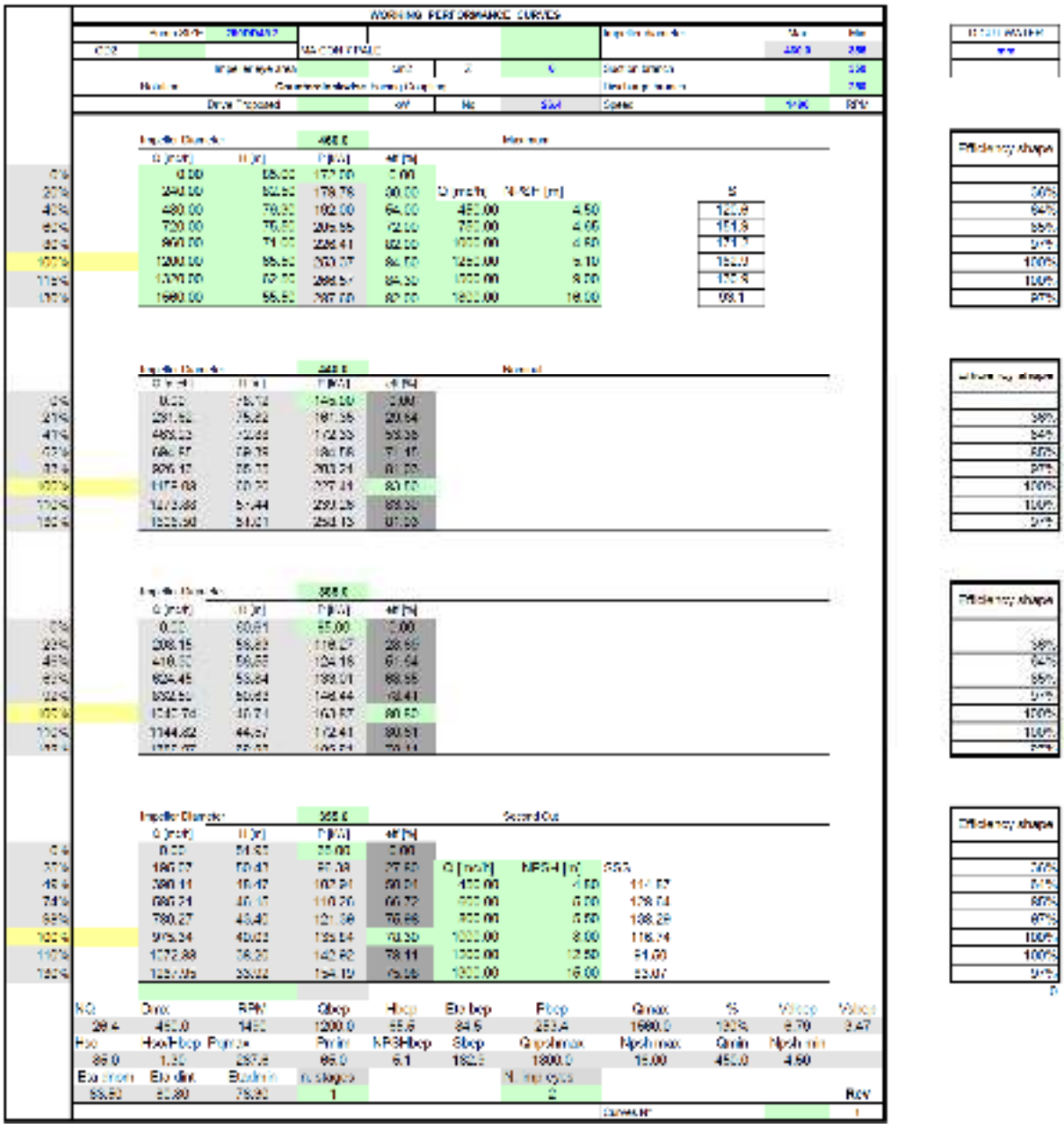

Figure 1. Molude 471.

4.2. PROCESS: SCALING

Speed Scaling

Fitting is governed by 4 exponents to manage efficiency and NPSH3, each at reduced and increased speeds. With this approach there is no need to produce different curves. Exponents are calculated by affinity laws and verified with tests, if available.

To generate curves at different speeds, one only must change the speed input in Molude 471.

Reference SOP-475 “Method for Refiguring Pump Performance from Tested Data” for more detailed information on speed scaling.

If a satisfactory fit is not achievable using the methods above, relating to such issues as exceeding the constraints of the intended tolerance grade, fit errors large enough to adversely impact the competitiveness of the final curves, or inadequately capturing particular characteristics of the hydraulic (e.g. head curve saddles), modifications to the approach can be made, including: overriding values of the calculated trim curves with those of curves fitted directly to test data, the use of additional curve points, use of multiple interpolated spline segments, or the use of other calculation tools to suit the particular task, being mindful that the best practice is to use the minimum possible number of points and spline segments.

Refer to the table in the previous trim curve section for acceptable curve fit errors under various conditions.

Geometric Scaling

Reference SOP-475 “Method for Refiguring Pump Performance from Tested Data” for procedures for geometric scaling. Molude 471 provides the functionality to perform this scaling in accordance with SOP-475.

The calculated efficiency can be overwritten according to test data with two factors: one applied over the entire flow range, and one increasing with flow, to replicate friction losses (refer to example from Molude 471 below).

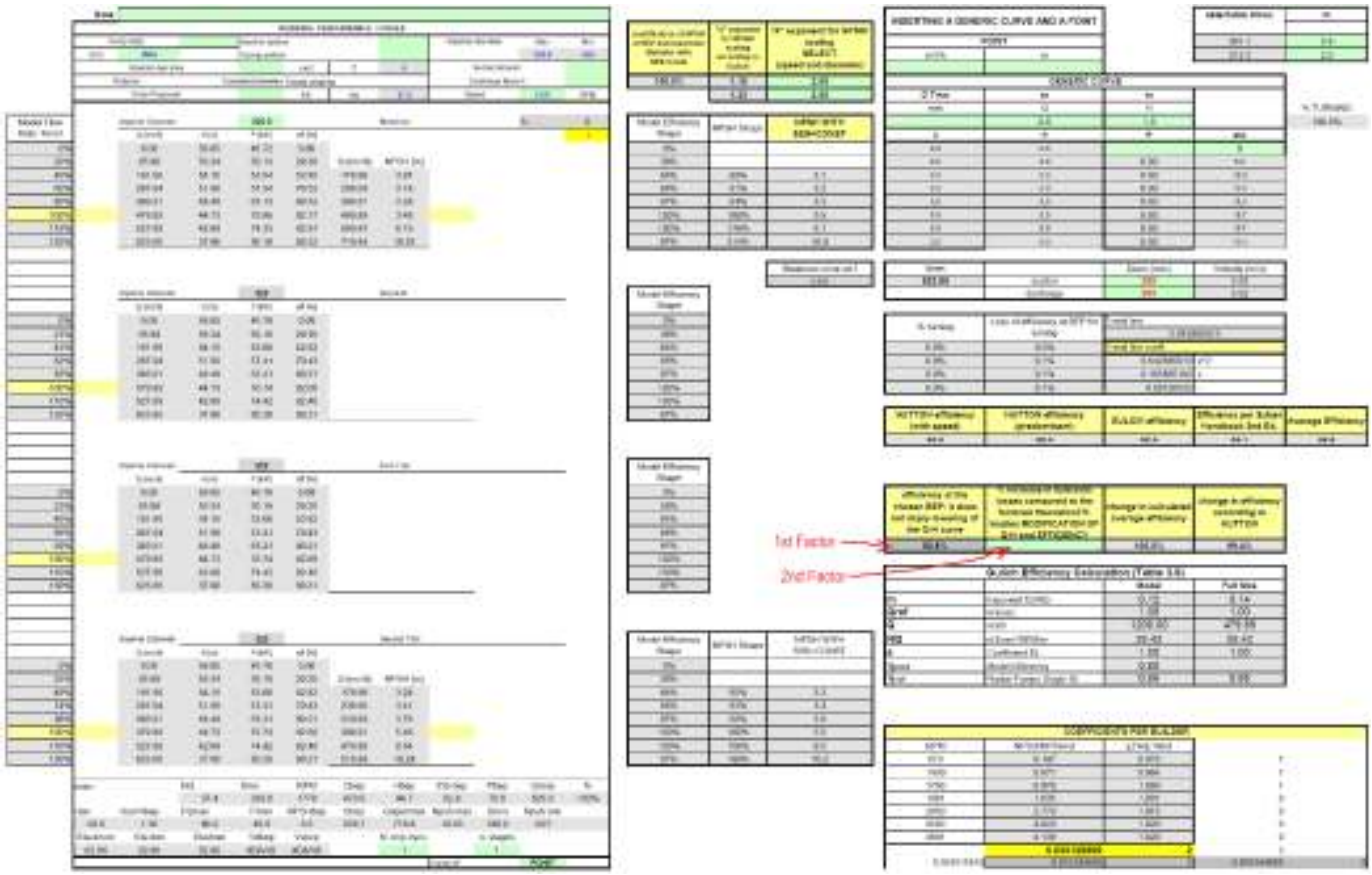

Figure 2. Efficiency adjustment for geometric scaling in Molude 471.

4.3. PROCESS: INSTABILITY

Instabilities

Instabilities are differentiated as either:

1) those where (starting from zero flow) head increases then decreases (sometimes referred to as a curve "droop" at shutoff), or
2) those where head decreases then increases (ie, a "saddle" type of instability). The first is typical of low specific speed, the second of high specific speed, especially for vertical pumps.

The first type of instability is usually shown in the curve, with the exceptions typically made where API 610 Std permits use of an orifice for continuously rising head, or for horizontal multistage pumps not tested at shutoff. For these cases, the first instability is shown as a horizontal Q-H contour between zero flow and Q(max Head). Q (max Head) is set as MCSF.

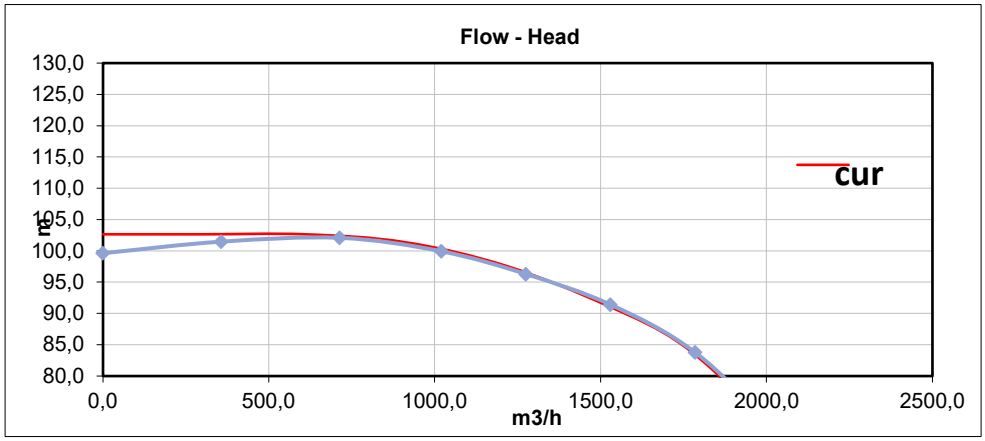

Figure 3. Instability at shutoff

Test Tolerances according to different standards, for reference:

			API		HI (only rated point)				
	guarantee requirement		HSO tolerance	Rated point	1B	1E	2B	1U	2U
	ft	m	%	%	%	%	%	%	%
HEAD	0 - 250	0-75	±10	±3	±3	±4	±5	0 +6	0 +10
	> 250 - 1000	>75-300	±8						
	>1000	>300	±5						
FLOW			NA	NA	±5	±5	±8	0 +10	0 +16
POWER	optional (either/or)		NA	+4	+4	+4	+8	+10	+16
Efficiency			NA	NA	-3	-0	-5	-0	-5

Table 3. Test tolerances according to industry standards.

Adjustment of HSO according to product line (pump features) and test tolerance:

PL / HEAD	test data max adjustment VS max adjustment			
	0 - 250 ft	> 250 - 1000 ft	>1000 ft	
	0-75 m	>75-300 m	>300 m	Note
OH2 w Orifice	±4	±3	±2	Orifice and trim allow a great flex to achieve tolerance
OH2 w/o Orifice	±3	±2	±1,5	
VS4 w Orifice	±4	±3	±2	Orifice and trim allow a great flex to achieve tolerance
VS4 w/o Orifice	±3	±2	±1,5	
BB1	±3	±2	±1,5	
BB2 w Orifice	±4	±3	±2	Orifice and trim allow a great flex to achieve tolerance
BB2 w/o Orifice	±3	±2	±1,5	
BB3/4/5	±2,5			regardless Head since Shutoff is no possible to test or test data are not a real measure
Other VS	N/A			

Table 4. Adjustments to HSO.

The second instability is always shown (refer to example below).

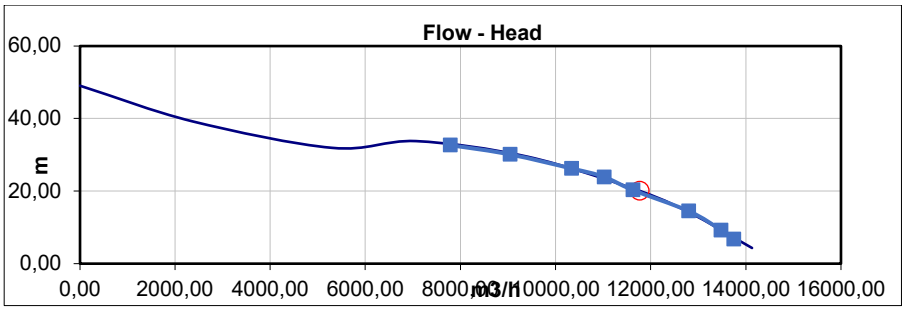

Figure 4. Saddle type instability.

4.4. BEP DEFINITION

A polynomial curve of 4th order is used to interpolate the efficiency curve and define the BEP position. This functionality is built in for both Molude 471 and the SKB.

4.5. END OF CURVE DEFINITION

Fitted curves for upload to IQ are will generally be defined to the maximum flows indicated by testing. Among the reasons for this is to ensure adequate motor sizing where the power curve rises to runout and to ensure that the data is available if required by the job. The end of curve that is displayed in IQ is a separate option that can be tailored to individual project requirements.

The end of curve is a separate value that limits the AOR. If using Molude 471, default values shown are 130% or 120% but can be managed during the selection up to the maximum values allowed. The following is a guide for EOC values by pump type:

Pump Type, Application (if relevant)	Max Allowable flow	Default shown EOC (IQ)
VS1/2, VS6/7	Varies by model, based on test. Generally 130% or more.	130%
VS4	130%	120%
OH2	130%	120%
OH2 low flow	110-125%	110-125%
BB 2 - 3 - 4 -5	130%	120%
BB1	Varies by model, based on test. Generally 130% or more.	120%
OH6 alternative	Varies by model, based on test.	Varies by model, based on test.

Table 5. Typical EOC values (% of BEP flow).

When generating curves at multiple trims or speeds that are not calculated by affinity but rather fit directly to test data, the end of curve should be coordinated among the related curves based on affinity.

IQ can provide an option to have a default EOC of 120% or 130% BEP, while retaining the possibility to extend it if required by an application or client.

4.6. MCSF

Refer to SOP-461 “MCSF Method and Calculation” for determination of MCSF. If test data is not available, calculation per SOP-461 can be used to establish the MCSF.

Generally, the MCSF percentage increases as specific speed increases.

For product lines that must meet API vibration requirements, the MCSF may be limited by a mechanical issue recorded during testing – refer to SOP-461 for additional information.

4.7. NPSHr

NPSHr is based on the NPSH3 test data and scaled for sizes having no data. A correction should be included in the curve for test and manufacturing variation as defined in the "Derates" section above. An additional margin is applied in IQ to insure the NPSHA for the application is adequate. The user can also increase NPSH3 in IQ to respect suction specific speed limitations required by the client.

Note: There is a margin warning in IQ for the duty point only, and an alert advises if NPSHa is less than NPSH3 at EOC. Different margins over the POR/AOR are not included in IQ, but especially for high energy applications, there is a dedicated minimum margins calculation available separately.

Specific NPSH margins applied in IQ are outside the scope of this document.

Reference SOP-475 "Method for Refiguring Pump Performance from Tested Data" for procedures for geometric scaling of NPSHr.

4.8. FLOWCHART

The following is the curve fitting procedure for a new hydraulics with data from CFD or test.

The flowchart describes the first data entry for a new hydraulic or a new data fitting.

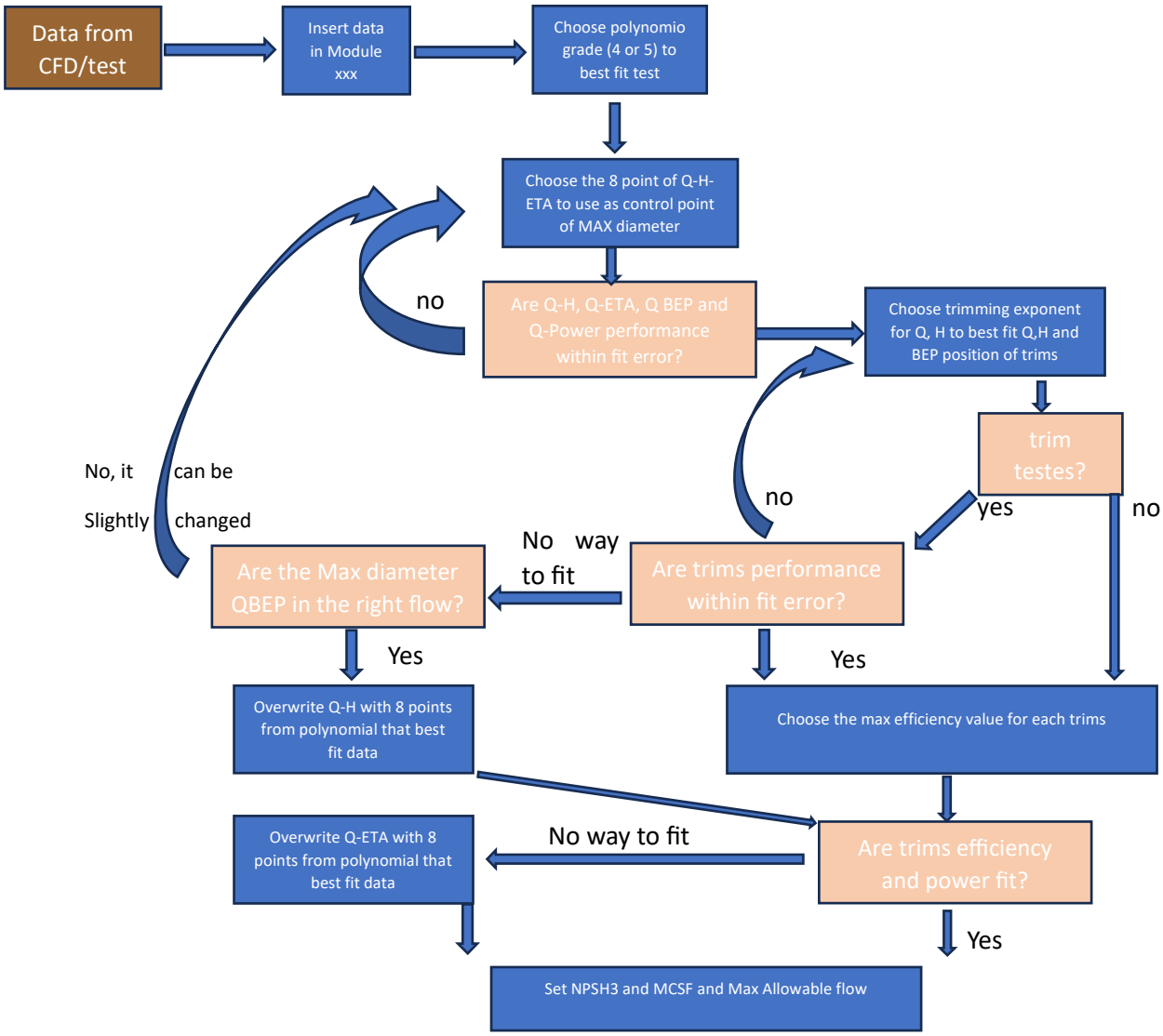

Figure 5. Flowchart of the curve fitting process for new hydraulics.

For creating a scaled size please reference the existing sheet where the test data is fitted and check the above steps. Once the first sheet is completed, a "standard sheet" must be duplicated with the exact size we are going to publish. If we want to publish exactly the size in the test module we only need to add the same max diameter in the "standard sheet" and insert the desired speed, curve number, exponent for trimming Q, H, and scaling NPSH3, and max efficiency values.

In this way we have a “scaled” size (even if the scale is 1.0000), managed as a standard sheet to export to IQ with all needed outputs:

1. Set of Q-H, Q-ETA for a maximum of 4 trims.
2. Set of exponents for NPSH and efficiency for increasing and decreasing speed.
3. Pump curve number.
4. Pump name and associated pattern.

5 DATA FROM TEST

Some test data may require some initial cleanup. Examples would be sets of data made up of multiple test runs or tests where multiple readings were taken at each flow. In these cases, an initial polynomial fit may be performed, and curve points defined from the fit at desired flows for further processing in Molude 471 or another tool. The scatter of the test data should be noted for later evaluation against the intended test tolerances, along with any applicable derates. For unilateral tolerances such as HI 1U, derates must be defined such that risk of test failure is eliminated for the observed test scatter and curve fit errors (refer to Sections 5.1 and 5.2 above for limitations defined for the latter). A useful verification practice is to plot the final fitted curve over the original test data to ensure that it is conservative.

6 GUIDELINES FOR SKB TOOLS

The curve fit is performed on efficiency. A single power value is defined at shutoff if needed.

Use 8 points minimum for fitting head and efficiency curves. Flow points shall include at a minimum first 0%Q, 15-35%, 35-65%, 65-75%, 75-85%, BEP, 110-120, 130% (or if available up to 150%) of BEP.

Use 6 points minimum for NPSH curves.

The minimum number of points should be used that enables an adequate fit. Unnecessary points only increase the difficulty of fitting, but additional points may be included as necessary to adequately capture anomalies such as saddles in the head curve. Extra points not used for fitting can be disabled in the SKB.

The use of a single polynomial for the entire curve is most desirable as it will yield the best appearance, provided a satisfactory fit in terms of deviations from the underlying curve data can be achieved. In more difficult cases (e.g., head curve saddles or other unusual curve shapes) division points (multiple polynomials) and slope overrides may be used as needed to improve the fit. However, an overly unconstrained fit as evidenced by false slope reversals, or slope reversals driven by random fluctuation in the underlying data, must be avoided.

In addition to acceptable appearance, SKB error values are used to evaluate the fits. Errors should be minimized and not exceed the standard governing test tolerances for the product line with allowance for expected test scatter. Allowance for test scatter is +/- 0.25% if it cannot be determined from available test data. Additionally, for consideration of unilateral test tolerances, negative valued errors for head and efficiency and positive valued errors for NPSHr may be unconservative and should be less than any derates or margins applied to account for manufacturing and testing variability (such derates shall be applied separately in IQ according to

the selected test tolerance). Individual test points that deviate excessively from adjacent curve points may be omitted if they are suspect.

7 UNITS OF MEASUREMENT

Input units can be US or metric as IQ can display either, but units for the fitted curves shall be consistent within product lines.

8 REFERENCES

Mod. 471: Best fit curves generator and hydraulic tests storage

SOP-461: MCSF Method and Calculation

SOP-475: Method for Refiguring Pump Performance from Tested Data

SOP-483: Method for Refiguring Pump Performance with Surface Finish

SOP-485: Method for Refiguring Pump Performance from Clearance Change